

Multi-Subject *Human Activity Recognition*

Ambient Intelligence and Domotics

Master Degree: Artificial Intelligence for Science and Technology

Gabriele Civitarese

EveryWare Lab, Department of Computer Science

Università degli Studi di Milano

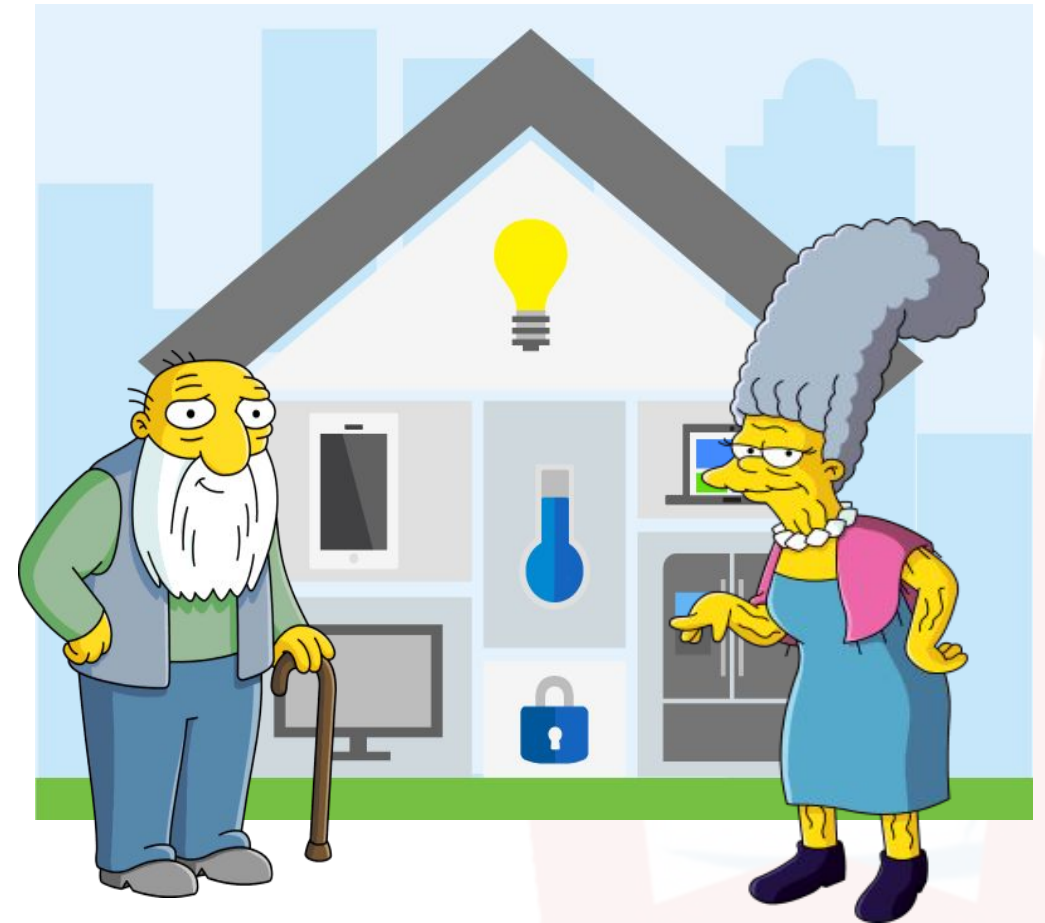
Copyright

These slides are from the teachers of this course. All the material is subject to copyright and cannot be redistributed without consent of the copyright holder. The same holds for audio and video-recordings of the classes of this course.

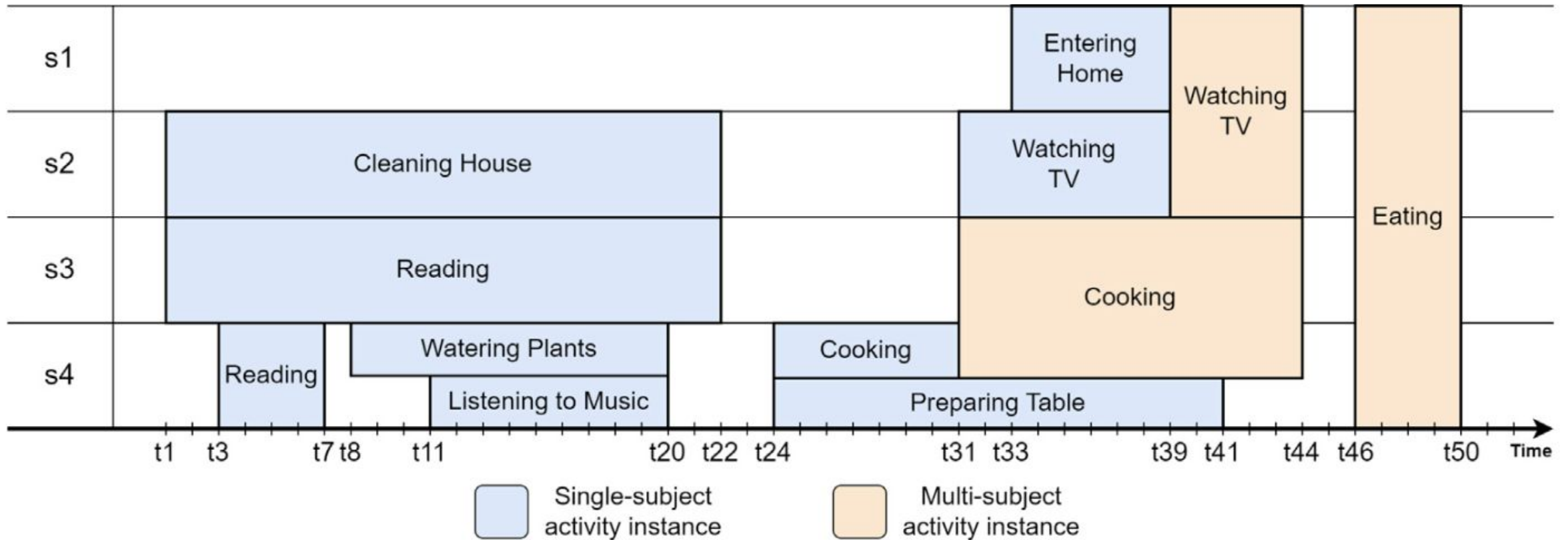
Multi-Inhabitant HAR in Smart Indoor Environments

HAR in smart environments: the Multi-inhabitant setting

- The majority of works in the literature focuses on activities of daily living (ADLs) in **single-inhabitant** settings
- Multiple subjects may perform:
 - the same activity in **cooperation** (e.g., Bob and Alice are cooking together)
 - individual activities in **parallel** (e.g., Alice is watching tv while Bob is watering plants)
 - group/individual activities in **parallel** (e.g., Alice and Bob are cooking, Carl and Dave are watching TV, Erica is reading)



An example of multi-subject activities



Which applications?

- Besides smart-homes and healthcare applications, there may be many other applications in other smart environments benefiting from multi-subject HAR, for instance:
 - office automation (e.g., identifying business meetings, presentations, etcetera)
 - building management systems
 - surveillance/security
 - work activities
 - smart retail (e.g., tracking behavior of multiple customers)
 - sport

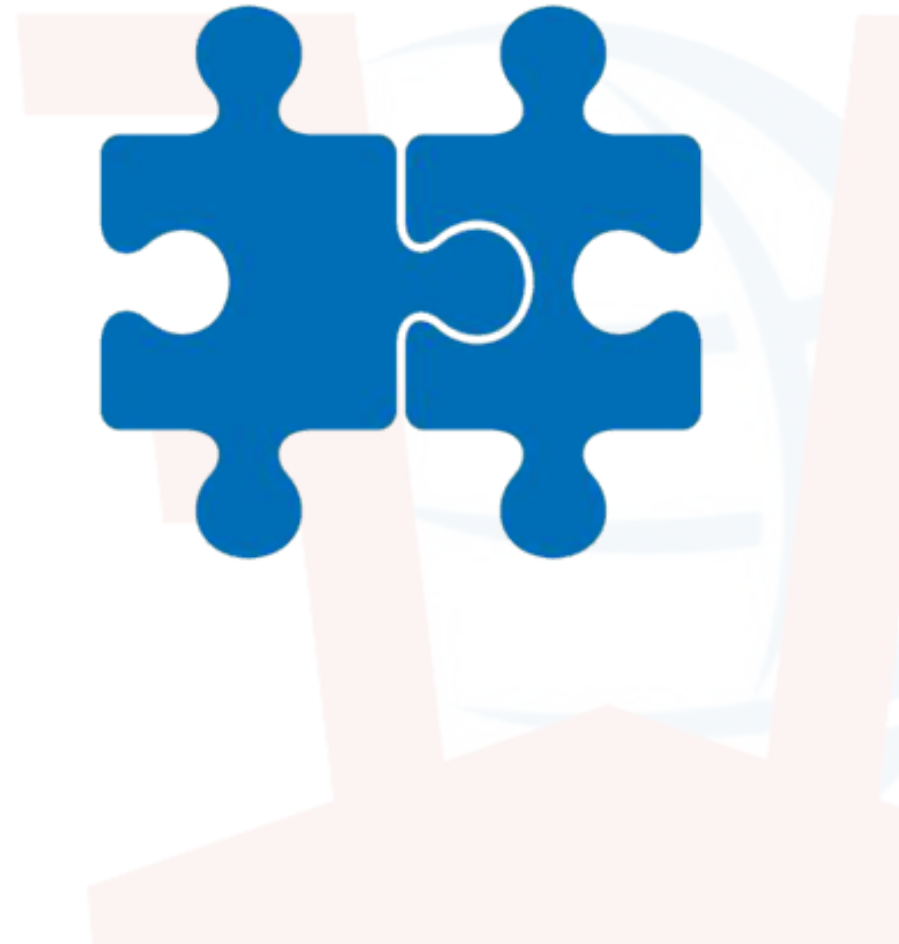
A brief recap on sensing

- As we discussed previously in this course:
 - camera-based solutions would be great in recognizing activities from multiple subjects, but they may be perceived as too privacy intrusive
 - wearable/mobile devices alone can detect physical activities, but they can not capture complex Activities of Daily Living (ADLs)
 - environmental sensors can unobtrusively capture the interaction of the residents with the home infrastructure, revealing the execution of ADLs
- In this lecture, we will mainly focus on how to rely on environmental sensors deployed in the environment to capture the ADLs performed by multiple subjects

Data Association

The “Data Association” problem

- Environmental sensor events (e.g., the stove is turned on, the fridge door has been opened) **can not identify** the inhabitants who triggered them
- This is not a problem in single-resident settings, but definitely when two or more residents are in the home
- How to correctly **associate** each environmental **sensor event** to the **inhabitant** that triggered it?
- One of the most challenging problems limiting the deployment of real-world HAR systems in smart-homes



Data Association: an example

- Alice and Bob are in the kitchen
- Alice is relaxing at the dining table while Bob is cooking
- A pressure sensor on the chair senses that someone is sitting
- A magnetic sensor senses that someone opened the fridge
- **How to correctly associate each sensor event to the corresponding inhabitant?**



Multi-inhabitant annotated sensor log

sensor_id	sensor_status	ts	subject_id
R5	ON	1542732888280	203
R5	OFF	1542732891359	203
R7	ON	1542732896144	203
R7	OFF	1542732899694	203
E1	ON	1542732913858	203
R2	ON	1542732919013	209
R2	OFF	1542732922270	209
R7	ON	1542732925166	203

In this example, we can observe that each sensor event is annotated with the resident triggered id ([MARBLE](#) dataset)

Multi-inhabitant annotated sensor log (cont.)

Timestamp	Sensor	Value	Annotated Activity
2009-08-24 20:23:43.054643	Kitchen_Stove	ON	R2_Prepare_Dinner begin
2009-08-24 20:23:56.086543	WorkArea1_Computer	ON	R1_Work_In_Room begin
2009-08-24 20:23:57.002937	Kitchen_Fridge	ON	
2009-08-24 20:24:19.034964	WorkArea1	ON	
2009-08-24 20:24:21.061429	WorkArea1	ON	
2009-08-24 20:24:22.078563	Kitchen_Door	ON	
2009-08-24 20:24:23.077674	WorkArea1	ON	
2009-08-24 20:24:25.095411	BedArea1	ON	
2009-08-24 20:24:26.077674	WorkArea1	ON	R1_Work_In_Room end

- Due to the challenges related to acquiring labeled datasets, it is often prohibitive to annotate, for each sensor event, the resident that triggered it
- In this example, we see two parallel activities performed by residents R1 and R2 ([CASAS](#) dataset)
- This is the most likely annotation scenario

Data-driven Data Association

Towards data-driven data association

- Data association could be tackled as a data-driven problem
- Goal: training a classifier that associates each sensor event to the resident that actually triggered it
 - then, it may be possible to use a classic single-subject HAR model for each subject

A naive Probabilistic Approach

Idea: using a probabilistic method (Naive Bayes) to estimate the probability that a given sensor event was triggered by a resident:

$$\text{Likelihood}(\text{Event}_i \text{ triggered by Resident}_n) = P(\text{Resident}_n)P(\text{Event}_i | \text{Resident}_n)$$

- $P(\text{Resident}_n)$ is the probability of Resident_n of triggering events
- $P(\text{Event}_i | \text{Resident}_n)$ is the probability of observing a specific event for a specific resident
- Event_i is a complex feature vector encoding spatial and temporal relationships with past events and time features (e.g., time of the day, day of the week, etcetera)

[Crandall, A. S., & Cook, D. (2008, July). Attributing events to individuals in multi-inhabitant environments. In 2008 IET 4th International Conference on Intelligent Environments (pp. 1-8). IET.]

Drawbacks of data-driven data association

- The previous method could of course be extended with more sophisticated approaches
 - e.g., given a window, using LSTM models to analyze which events are triggered by which resident based on subject-specific patterns
- Drawbacks:
 - a significant amount of labeled data about relationships between specific subjects and sensor patterns is required
 - a trained data association model would be specific for a specific environment and set of subjects and can not easily generalize in different settings
 - ...is it possible at least to have a single end-to-end model for data association and HAR?

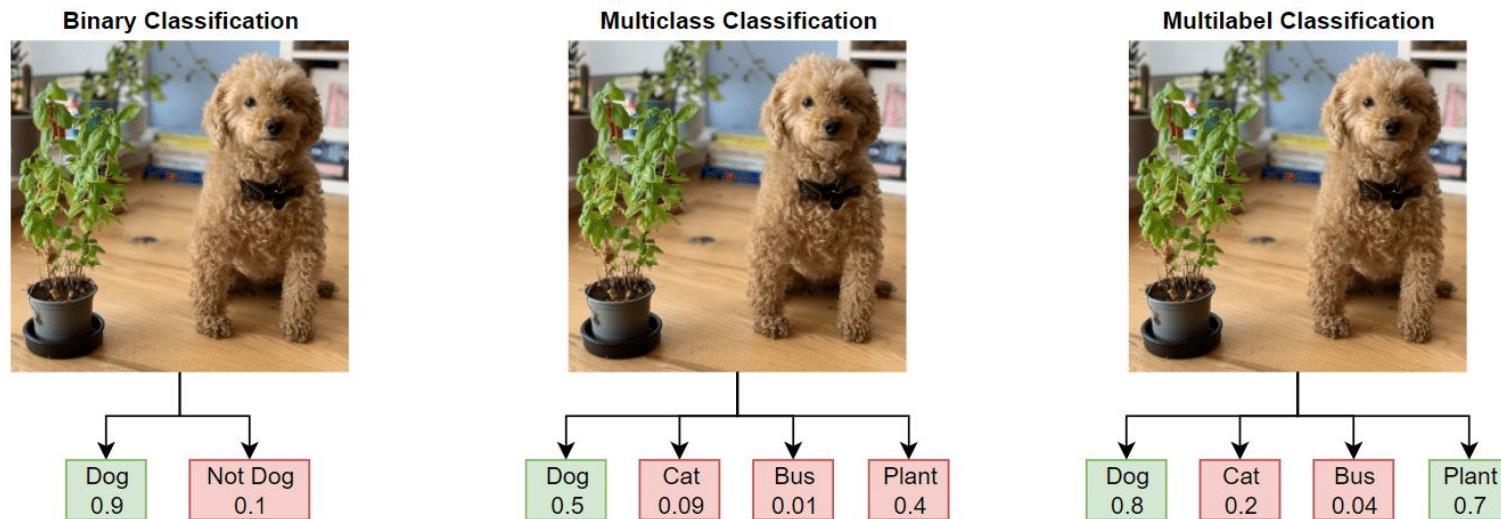
Implicit Data Association

“Implicit” data association

- Many methods do not consider data association as a distinct problem with respect to HAR
- These approaches rely on a single model
 - the input are windows of sensor events (where each window may reflect activities of multiple residents)
 - the output is the activity performed by each resident
 - this is may be achieved thanks to ***multi-label*** models!
- Hence, data association is performed “implicitly” by the model to infer the activities performed by each resident

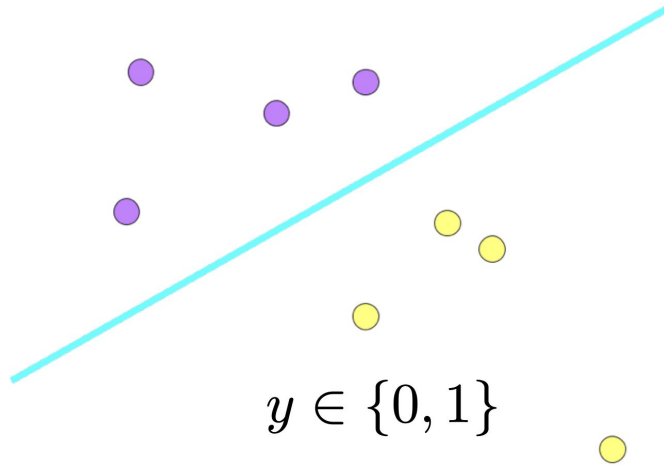
Preliminaries on Multi-Label Classification

- Multi-label classification refers to the setting where more than one output may be attributed to the same input
 - differently from multiclass, where only one label is associated to the input!

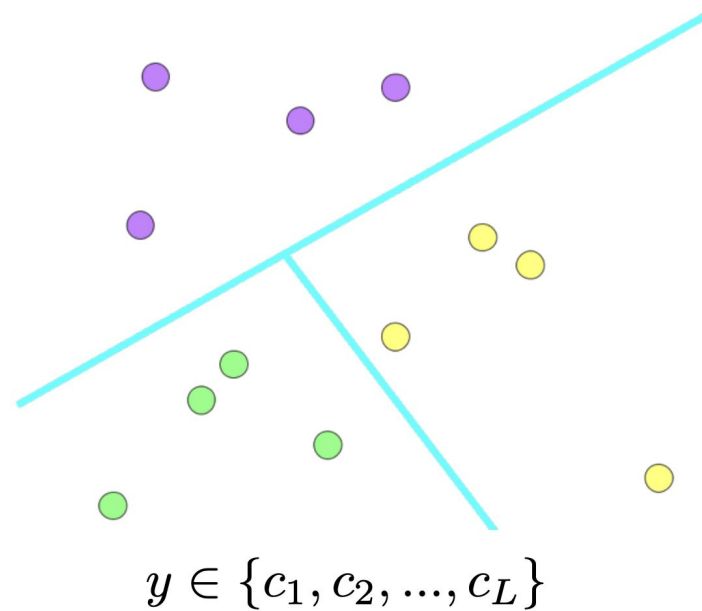


Binary VS Multiclass vs Multilabel

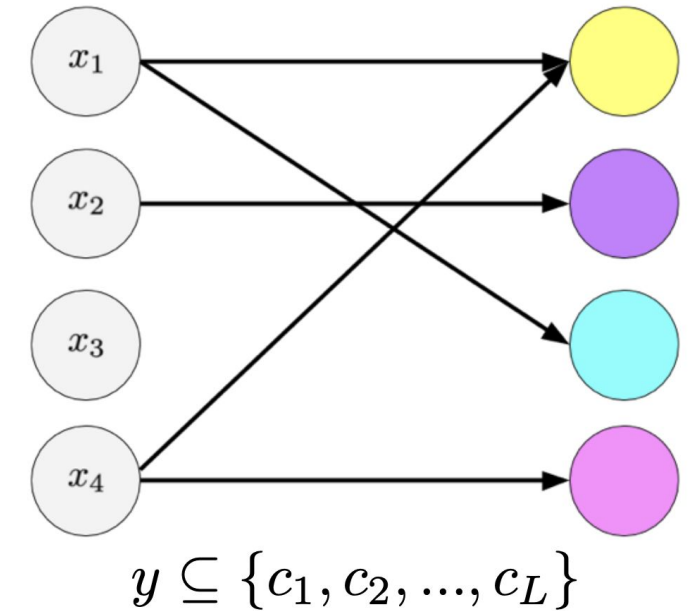
Binary Classification



Multiclass Classification



Multilabel Classification



Multi-label classification for multi-inhabitant settings

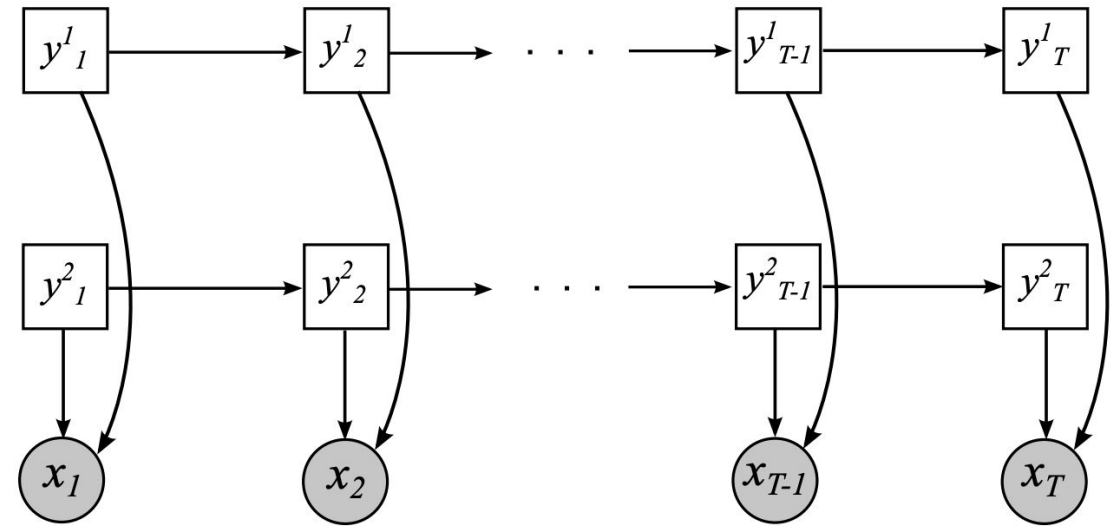
- In our case, the number of labels in output is fixed: one activity for each resident
- These labels are **not independent** between them
- In the following, we will go through some of the main approaches proposed in the literature to tackle this problem

Probabilistic Models

- Many research groups focused on graphical probabilistic models
 - probabilistic models with a graph structure
 - each node is a random variable
 - edges represent dependencies between random variables
- Goal: given a sequence of observable variables (sensor events), infer the values of hidden variables (activities) at each time step
- Several variations of Hidden Markov Models (HMMs) and other probabilistic models have been proposed for multi-inhabitant settings!
 - we will see only one example of such approaches

Method 1: Factorized HMM (FHMM)

- One independent probabilistic chain for each resident
 - observable variable x_t : sensor event at time t
 - hidden variable y_t^R : activity performed by resident R at time t
- Observable variables are the same for both chains!
 - thus introducing correlations between activities performed by the two subjects



[Alemdar, H., & Ersoy, C. (2017). Multi-resident activity tracking and recognition in smart environments. *Journal of Ambient Intelligence and Humanized Computing*, 8, 513-529.]

Goal of FHMM

FHMM aims at computing the following joint probability (in this case for two residents):

$$p(x_{1:T}, y_{1:T}^1, y_{1:T}^2) = \prod_{t=1}^T p(\mathbf{x}_t \mid y_t^1, y_t^2) p(y_t^1 \mid y_{t-1}^1) p(y_t^2 \mid y_{t-1}^2)$$

- Note that 1:T indicates the time sequence from the initial state (timestamp 1) to current time (timestamp T).
- Indeed, the goal is inferring *sequences* of hidden variables given *sequences* of observable variables

↑
probability of sensor
event X_t given the
activities performed by
the residents

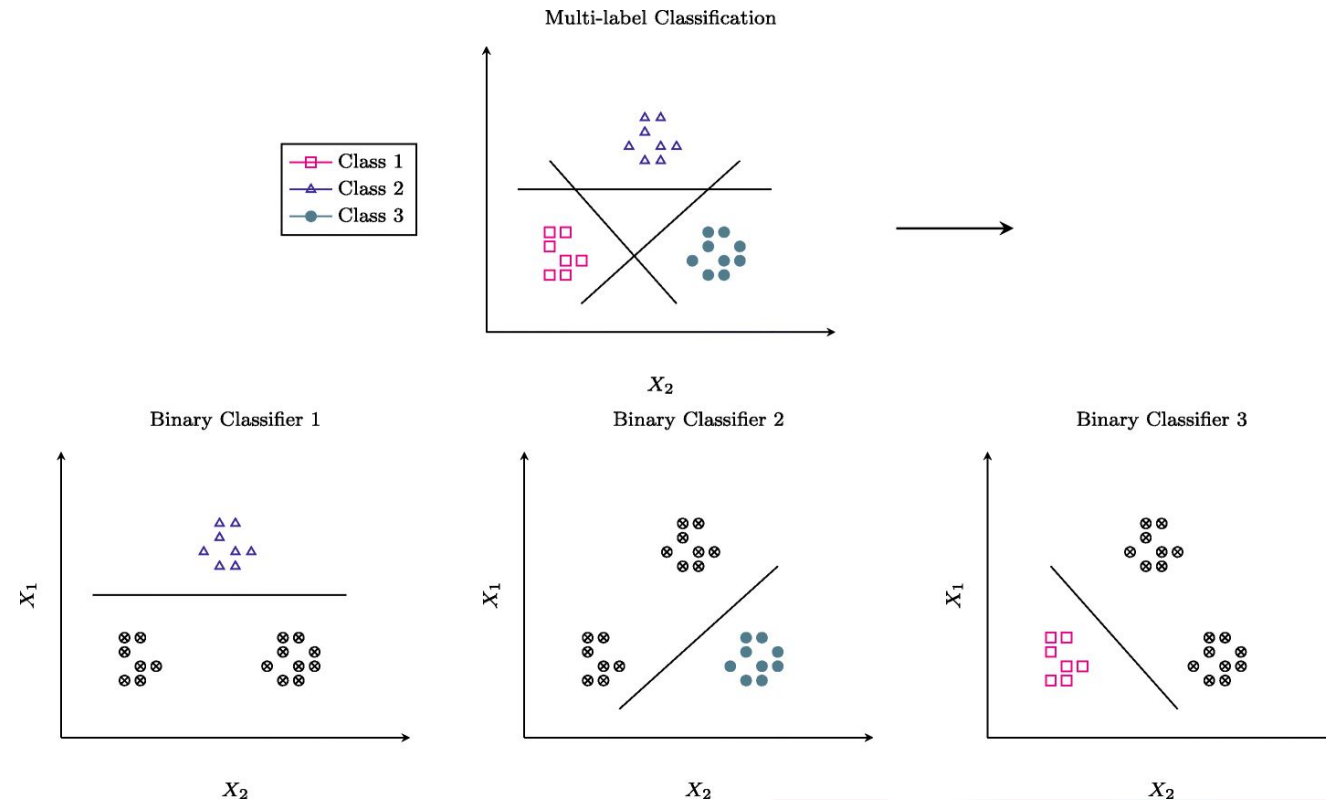
↖ ↗
transition probabilities
between activities
(for each resident)

Method 2: “Combined Labels”

- A simple solution leveraging single-label multiclass classifier by “combining labels”
- Idea:
 - suppose that there are three activities A1, A2, A3 and two residents R1, R2
 - a label is created for each possible combination of activities performed by the residents (e.g., A1A2 is the label identifying R1 performing A1 and R2 performing A2, A3A2 identifies R1 performing A3 and R2 performing A2, etcetera)
- Pro: very easy to implement
- Cons: with N residents and M labels, in the worst-case scenario this may result in N^M labels!!

Method 3: Binary Relevance

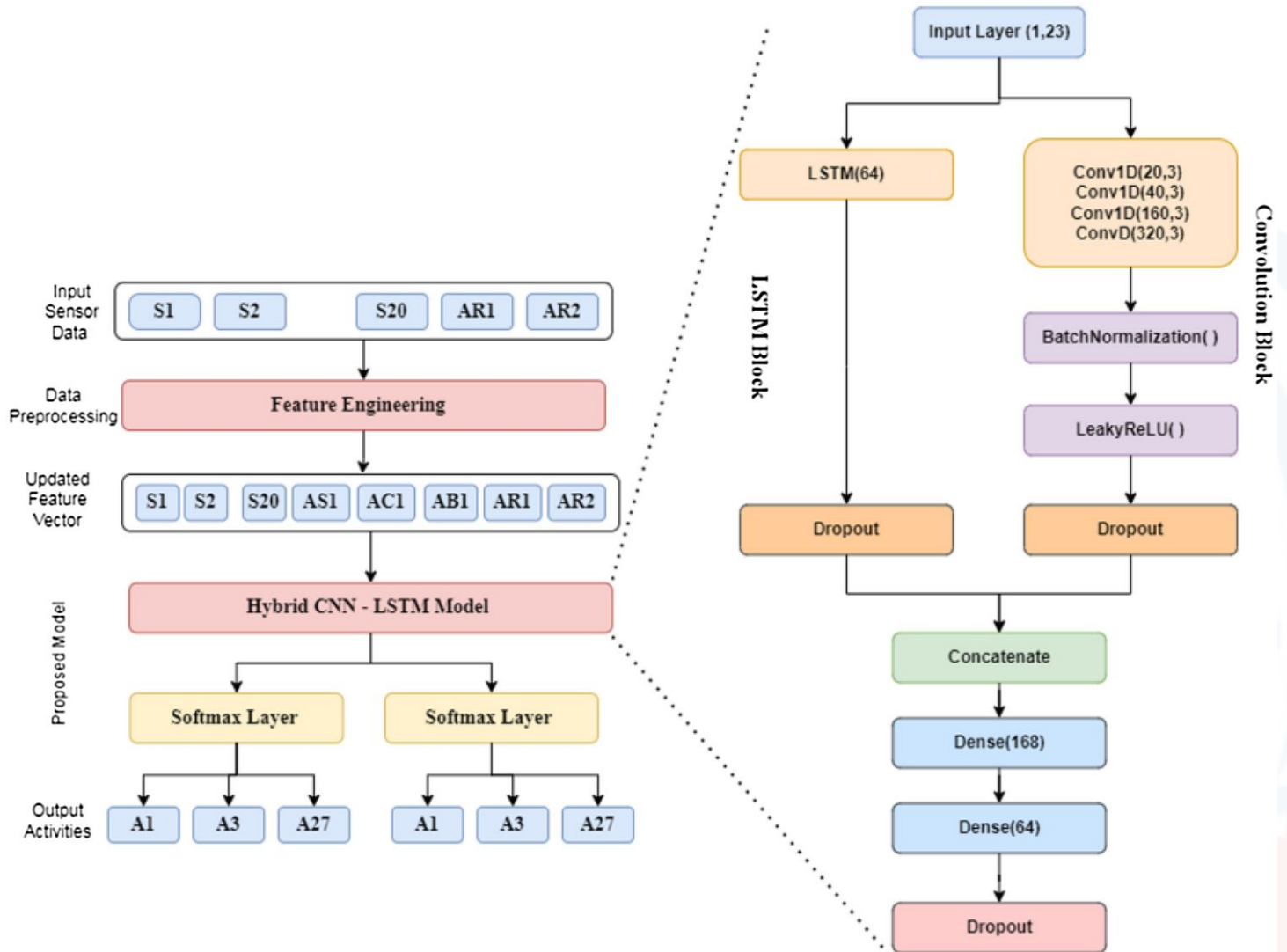
- A way to implement multi-label classifiers
- It requires building a binary classification model for each class
- An activity is provided as output if the confidence of the corresponding binary classifier is higher than a certain threshold
- **Drawback:** dependencies between the activities among multiple residents are ignored



Method 4: Multi-Output Deep Model

- An output layer (softmax) for each resident
 - in this case, two residents
- Shared layers makes it possible to capture spatio-temporal relationships between the activities performed (in parallel or jointly) by the different residents.

[Ramanujam, E., & Perumal, T. (2023). MLMO-HSM: Multi-label Multi-output Hybrid Sequential Model for multi-resident smart home activity recognition. Journal of Ambient Intelligence and Humanized Computing, 14(3), 2313-2325.]



Problems of multi-label approaches

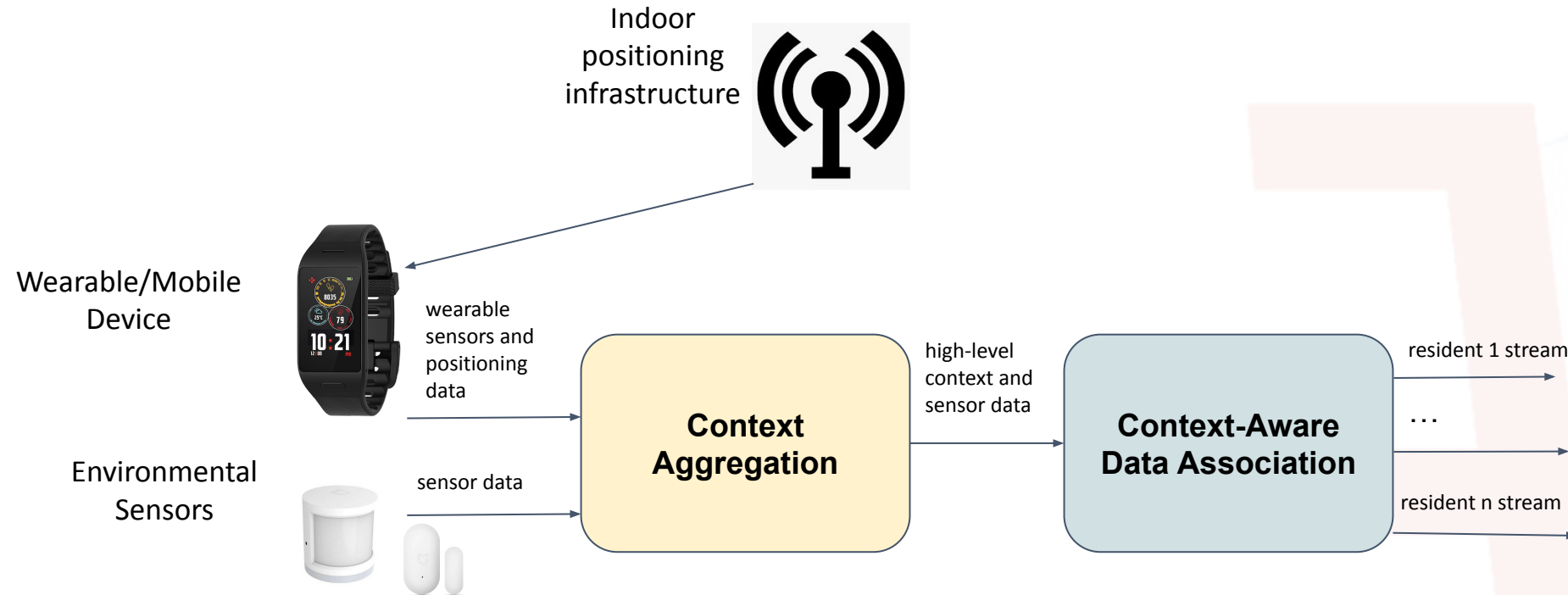
- Multi-label approaches require to collect a training set including all the possible combination of activities performed by multiple residents
 - not practical!
- Such models can not generalize well on different homes with different residents and environments
- These models also suffer from scalability problems
 - increasing the number of residents or the number of activities significantly complicates the training process!
 - some of them (e.g., probabilistic graphical models) are computationally prohibitive and can not scale well with more than 2 subjects

Context-aware data association

Using additional context to simplify data association

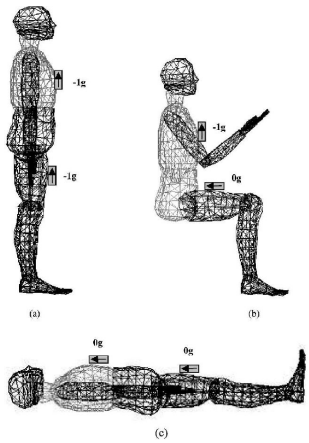
- Additional context information could be used to simplify data annotation
- For instance, assuming that each resident has a wearable device
- Each wearable device can track:
 - the identity of the resident wearing it
 - the location (e.g., if indoor localization is available)
 - low-level physical activities
 - etc.
- This context could be used to perform data association without training models!

Wearable-driven data association pipeline



Which contexts?

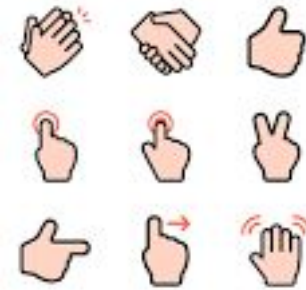
For each user wearing a mobile device, it is also possible to extract a wide variety of context information:



**User's
posture**



**User's
micro-localization**



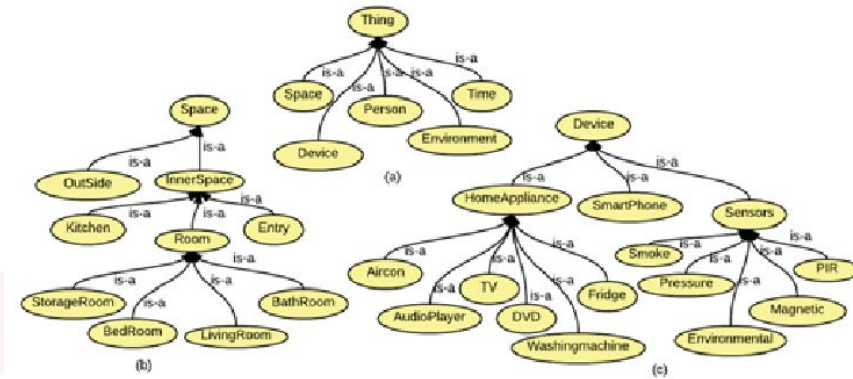
Gestures



Etcetera

Context-aware Data Association

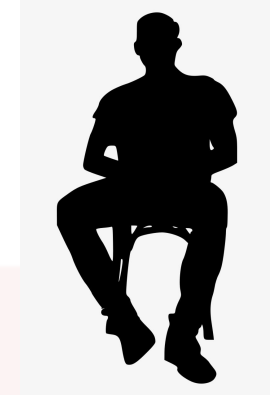
- Data association is based on a rule-based model defining **relationships** between environmental sensor events and context, based on common-sense knowledge
 - e.g., opening the fridge should be necessarily performed by the resident in the kitchen while standing
- When a new event is triggered, the knowledge can be leveraged to compute the **context-consistency** of each event with the context of each user u
 - i.e., context information (e.g., current location in the home, posture, gesture) are analyzed to understand if it is compatible with the sensor event
- A sensor event is associated with u only if context-consistent
- Advantage: performing data association without requiring training!



AN EXAMPLE

- **Events:**

- Fridge door opened (E1)
- Pressure sensor on the kitchen chair fired (E2)



- **Bob's context:**

- Standing
- In the kitchen



E1 is context-compatible,
while E2 is not

- **Alice's context**

- Sitting
- In the kitchen



E2 is context-compatible,
while E1 is not

Challenges of Context-Aware Data Association

- While context-aware data association does not require a learning phase, it has some drawbacks
- For instance, it is often impractical to assume that residents constantly wearing wearables, especially considering elderly subjects
 - this may be mitigated by miniaturized wearable devices such as smart clothes, smart rings, small BLE tags, etcetera
- The accuracy of data association is also constrained by the accuracy of collected context information
 - e.g., precision of micro-localization, detectable postures, gestures, physical activities

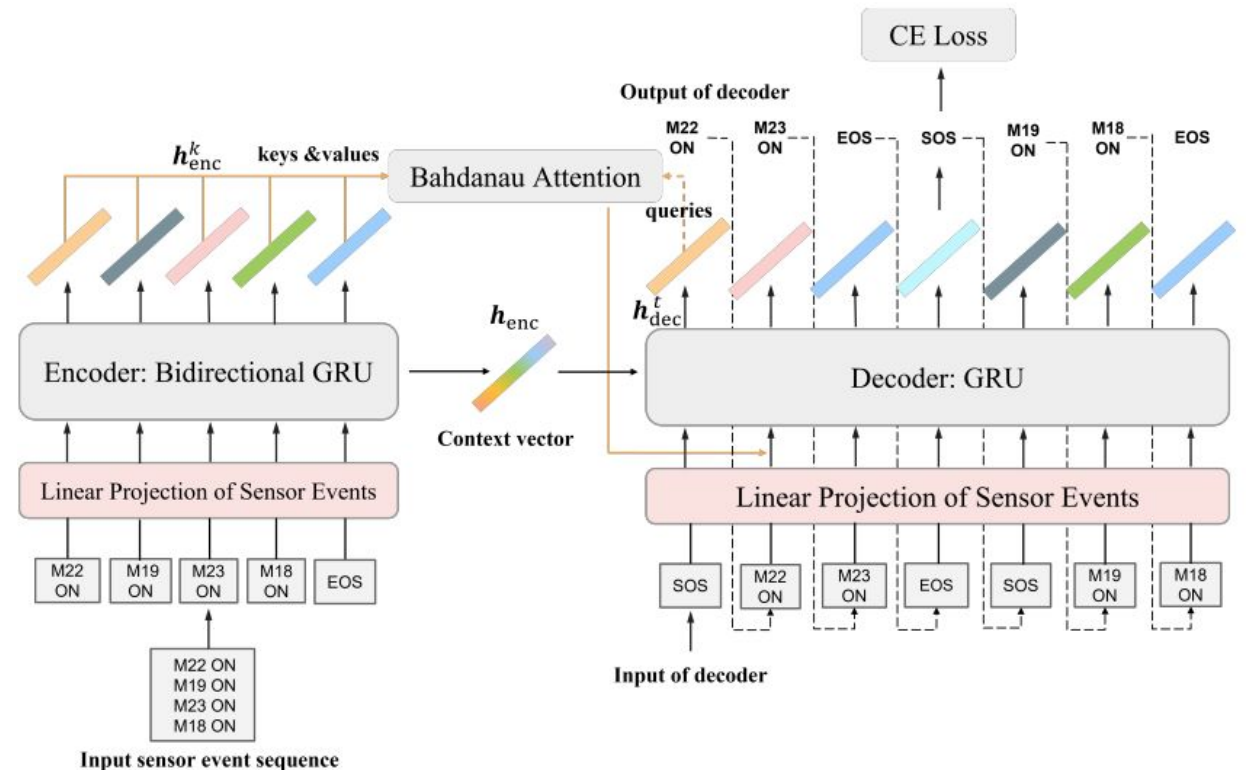
Subject Separation

Subject Separation

- For certain applications, it is just important to understand the performed activities without associating them with the corresponding resident
- In such cases, it is possible to perform a weaker form of data association called *subject separation*
- Idea: analyze the stream of sensor events just to determine which sensor events are more likely triggered by different subjects

Method 1: Supervised Subject Separation (seq2seq)

- Subject separation can be framed as a supervised seq2seq problem
- The **encoder** learns to map sequences of sensor events into a latent space
- The **decoder** learns to separate them into separate sensor sequences generated by different subjects
- Specific **tokens** (SOS, EOS) are used to separate these sequences
- **Drawback:** detailed annotation is still needed!!



[Generative Resident Separation and Multi-label Classification for Multi-person Activity Recognition Xi Chen, Julien Cumin, Fano Ramparany, Dominique Vaufreydaz, 2024]

Method 2: Unsupervised Subject Separation

- A method not requiring labeled data
- Assumption: in a first phase, it is possible to collect unlabeled data when only a single resident is at home
 - this may also be done automatically by detecting the presence of a single person in the home (e.g., based on PIR sensors patterns or on power consumption analysis)
 - of course, this assumption may not always be realistic
- Once unlabeled data are collected, statistics are extracted from single resident data
- Such statistics indicate the likelihood of sensor events being generated by the same subject

Unsupervised Subject Separation (cont.)

- The unlabeled dataset is hence mined to extract several types of statistics
 - e.g., frequency of sensor events pairs usually triggered consecutively, or temporally close
- Based on these statistics, it is possible to define thresholds to determine whether two sensor events are likely triggered by the same resident or not
 - e.g., only the couple of sensor events that occurred temporally close in the dataset at least K times are classified as performed by the same resident

An algorithm to mine temporally close sensor events

Algorithm *Temporally close events*(T):

Input: A temporal sequence of sensor events T ; a threshold ϵ (in seconds)

Output: A set R of records with associated statistics

```
1: for each couple of sensor events  $\langle se_1, se_2 \rangle \in T$  whose time distance  
   is less than  $\epsilon$  do  
2:    $c = \langle se_1, se_2 \rangle$   
3:    $c.occurrences = c.occurrences + 1$   
4:    $R.add(c)$   
5: end for  
6: for each  $c = \langle se_1, se_2 \rangle \in R$  do  
7:    $n_{se_1} = |\{se \in T : se = se_1\}|$   
8:    $c.frequency = \frac{c.occurrences}{n_{se_1}}$   
9: end for  
10: return  $R$ 
```

[Riboni, D., & Murru, F. (2020). Unsupervised recognition of multi-resident activities in smart-homes. *IEEE Access*, 8, 201985-201994.]

Method 3: Knowledge-based Subject Separation

- The common-sense knowledge of spatio-temporal relationships between sensor events may also be used to tackle *subject separation* as a knowledge-based approach
- Hence, it is possible to leverage *semantic and temporal similarity* between consecutive sensor events to determine whether they were triggered by the same resident or not (based on a threshold)

Semantic and temporal similarity

$$\text{sim}(se_i, se_j) = (\text{sim}_T(se_i, se_j), \text{sim}_S(se_i, se_j))$$

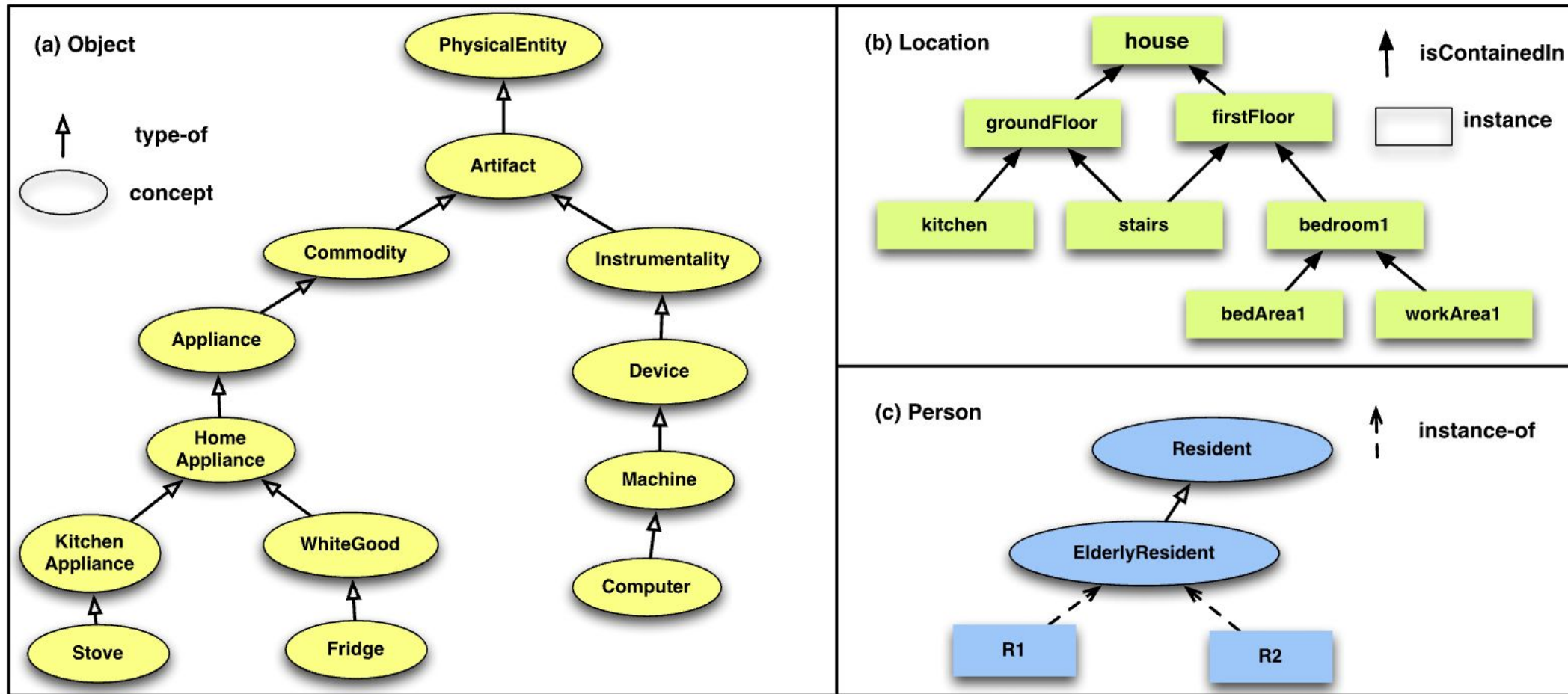
- Similarity computed on pairs of consecutive sensor events
- sim_T : temporal similarity
- sim_S : semantic similarity

Temporal similarity

$$sim_T(se_i, se_j) = \max \left(0, 1 - \frac{|\tau(se_j) - \tau(se_i)|}{T_{\max}} \right)$$

- $T_{\max}$ is the maximum range of the time under consideration.
- For instance, if we consider 24h, $T_{\max} = 84600s$

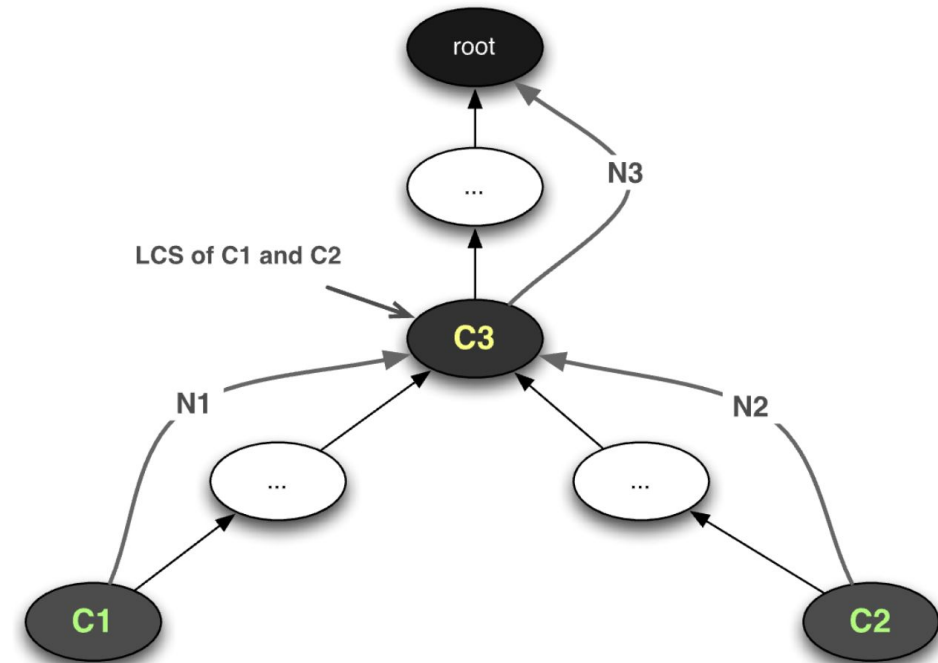
The knowledge-based model: ontologies



Concepts similarity

- In the ontology, the similarity between two concepts C1 and C2 can be computed finding the *Least Common Subsumer (LCS)*
 - *the most specific concept C3 that both concepts share as an ancestor*

$$sim_C(c_1, c_2) = \frac{2 \times N_3}{N_1 + N_2 + 2 \times N_3}$$



Semantic Similarity

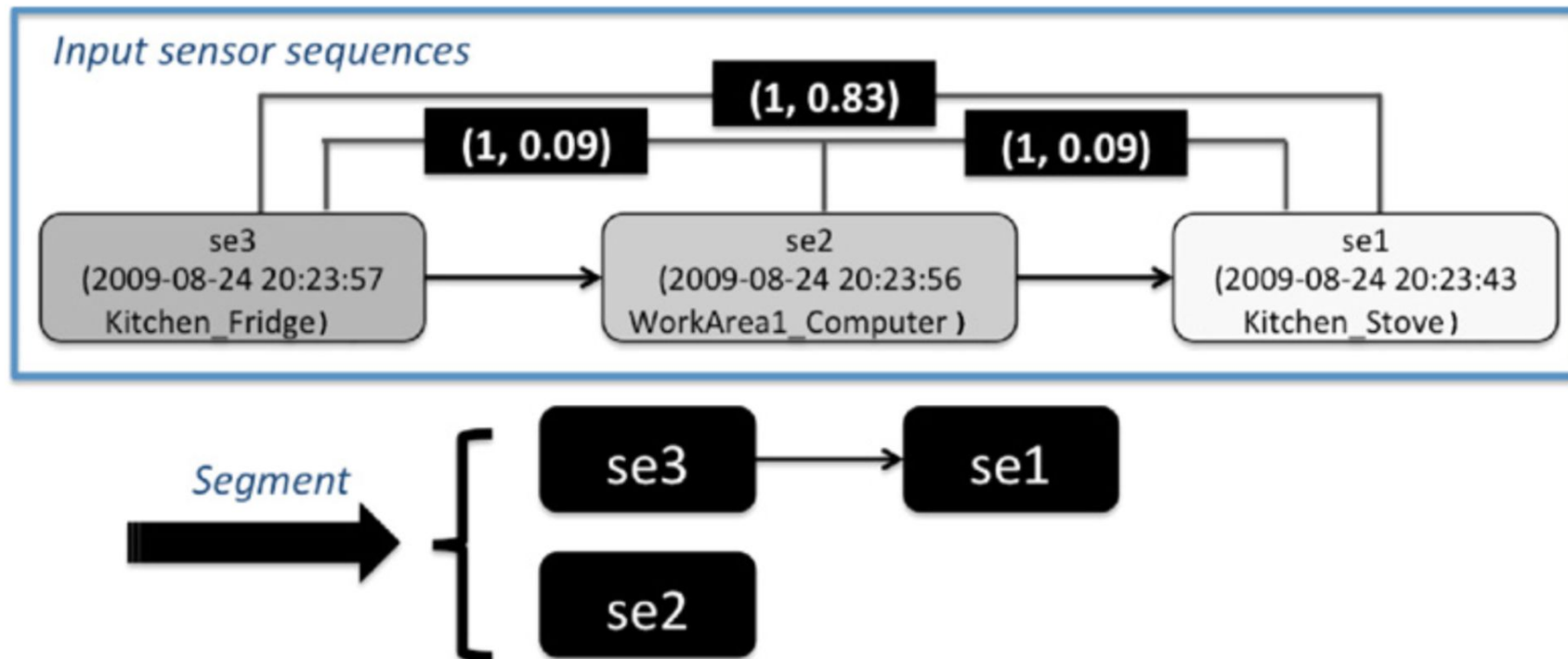
$$sim_S(se_i, se_j) = (sim_C(\iota(s_i), \iota(s_j)) + sim_C(\delta(s_i), \delta(s_j))) / 2$$

concept similarity in the
ontology

mapping sensor events to
their location

mapping sensor events to
their associated object

Deriving parallel segments based on semantic similarity



Challenges of subject separation

- It is still challenging to separate data of different residents if they are performing similar activities!
 - for instance, if Alice and Bob are both in the kitchen and performing activities involving similar sensors (e.g., Alice is setting up the table while Bob is cooking)
- Subject separation still may not scale well by increasing the number of residents

After data association/subject separation

Performing HAR

- Once *data association* or *subject separation* is completed, the output is a set of sensor streams (i.e., one for each resident)
 - each stream is also associated with the corresponding subject in case of *data association*
- At this point, it is possible to apply any type of single-resident HAR method on each stream individually!

How to detect group activities?

- Existing multi-inhabitant methods aim at recognizing the activity performed by each subject, but recognizing collaborative activities it is still an open challenge
- Possible assumption: if two or more residents are performing the same activity, they are also performing a group activity
 - however, this is true only in some scenarios
 - e.g., Alice is cooking, Bob is cooking, it is likely they are cooking together since there is usually only one kitchen in the home
 - in other cases, this may not be true
 - e.g., Alice is watching TV, Bob is watching TV but they are in two different rooms
- Context information should be considered in post-processing to determine if an activity is collaborative or not!

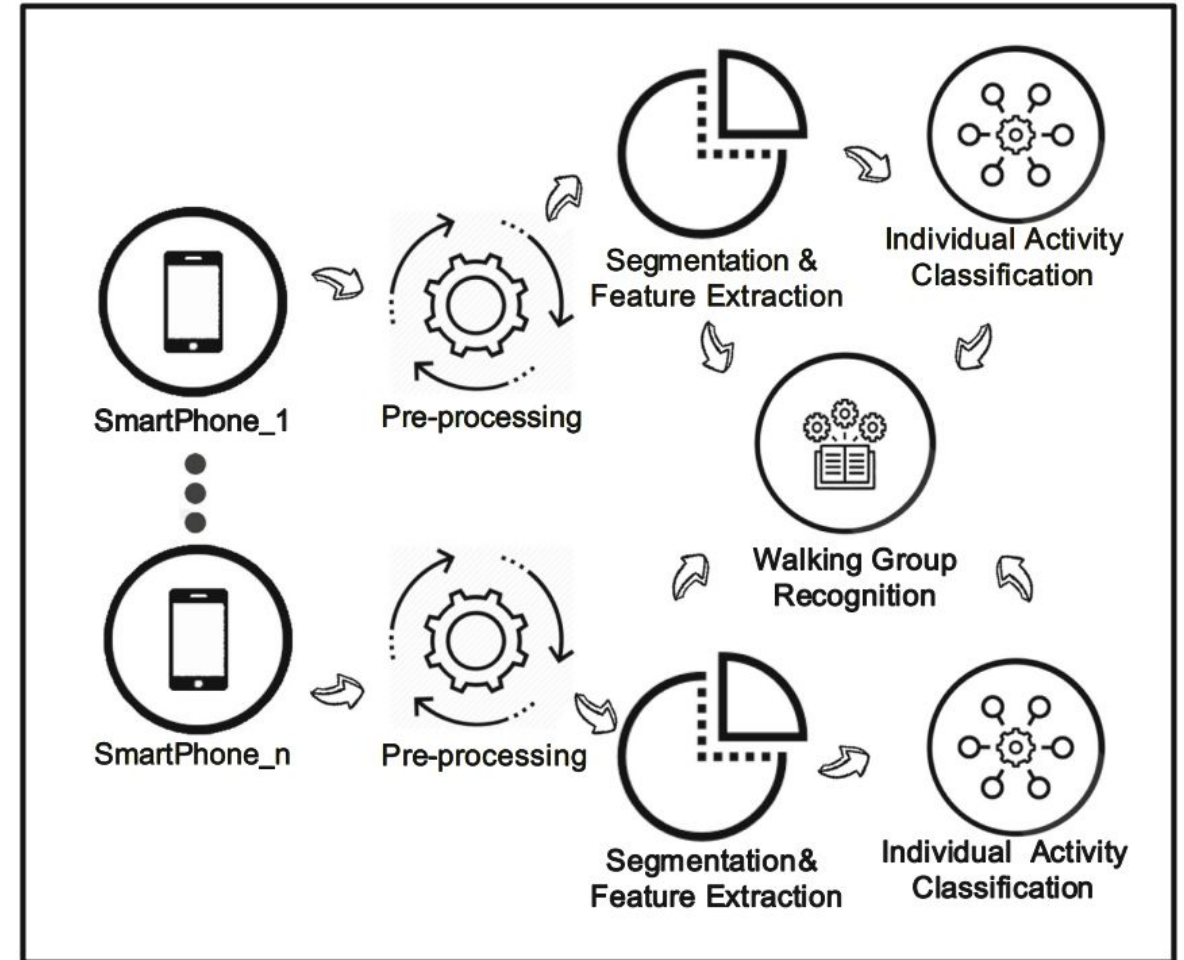
Multi-subject HAR only based on
mobile/wearable devices

Collective Behavior

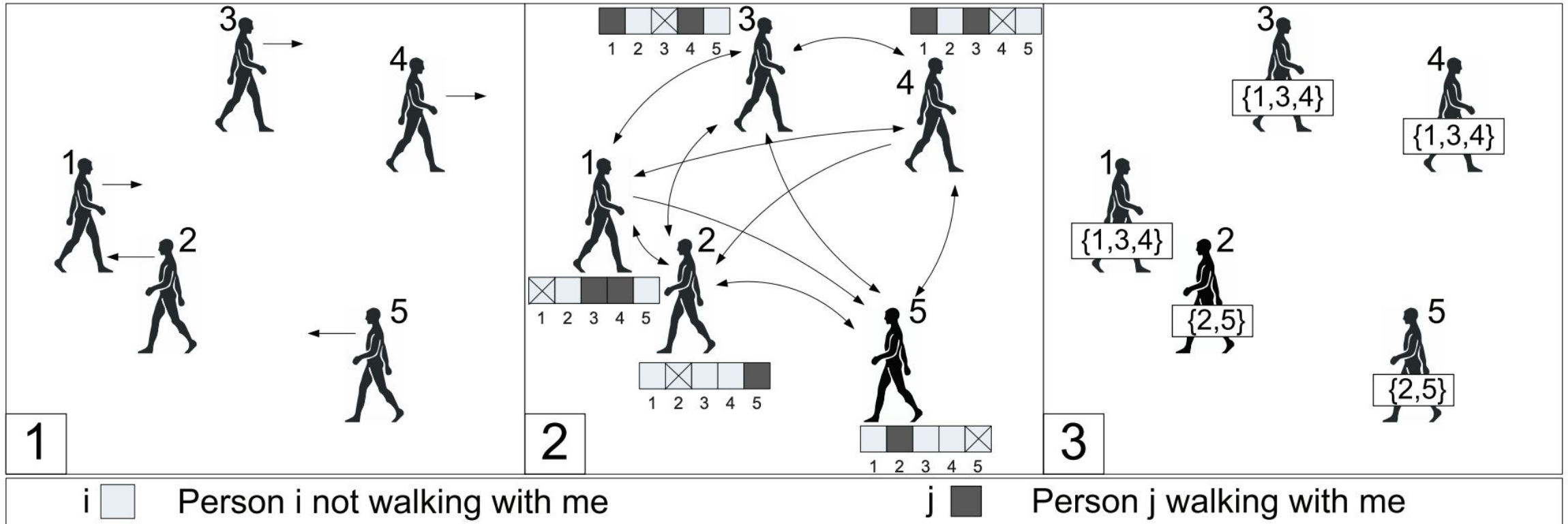
- By inferring the behavior of multiple subjects with mobile/wearable devices, it is possible to understand coordinated movements
- Examples:
 - queue of people
 - formation of unidirectional “lanes” in bi-directional pedestrian flows
 - the intersection lanes
 - group of people together at a specific location
- Applications:
 - emergencies (e.g., optimizing evacuation in large-scale public events)
 - smart city management (e.g., urban planning, optimized pedestrian navigation)

Distributed (centralized) approach

- Each device collects sensor data to:
 - recognize the most likely *individual* activity
 - e.g., accelerometer
 - contribute in understanding group activities
 - e.g., proximity with other subjects, direction
- A centralized server collects all the collected to recognize *individual* and *group* activities



A decentralized approach



Counting People in Smart Buildings

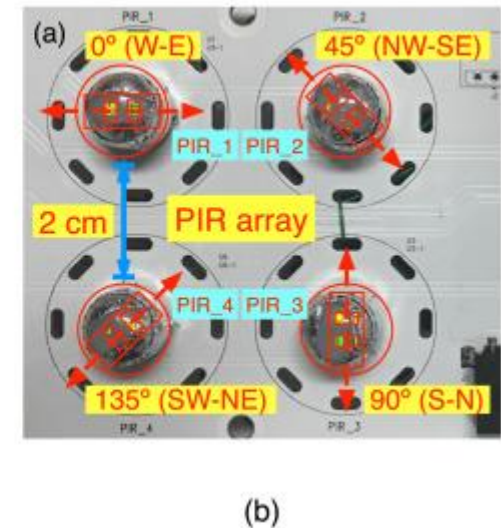
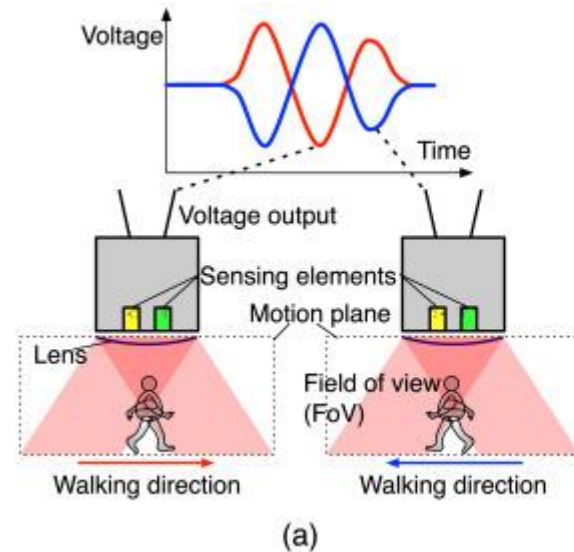
Counting People in Smart Buildings

- Sometimes it is not even important understanding the activities performed in a multi-subject environment, but just counting them!
 - For instance, behaviour patterns of occupants in using office spaces (e.g., such as office rooms, pantry areas, etc.), can be useful for energy saving
 - Heating, ventilation, air conditioning, lighting, and appliances could be adaptively and individually controlled per office space, using short-term (e.g., hourly) changes in occupant usage patterns.
- It is possible to estimate the number of people in specific indoor spaces using PIR sensors!



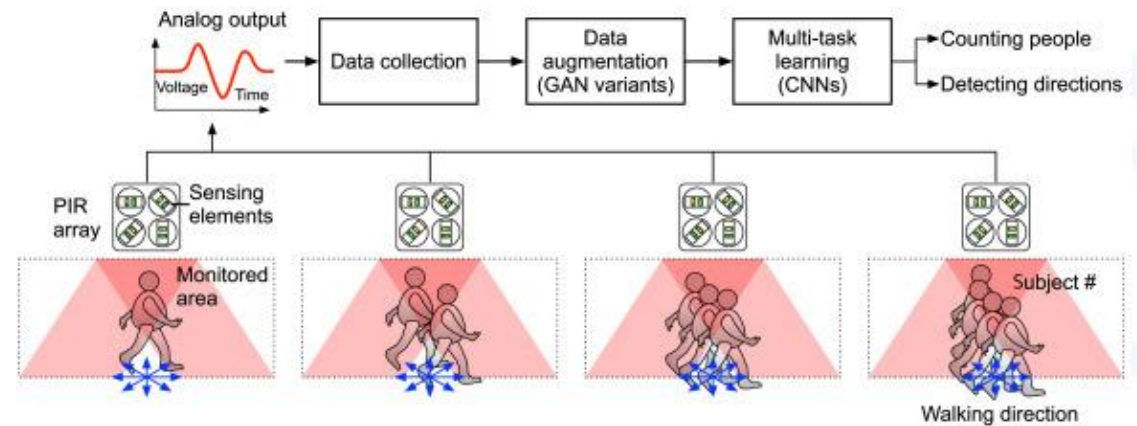
How to count people?

- Idea: using an **array** of PIR sensors pointing in different directions
- Hence, it is possible to infer the number subjects and their direction
 - e.g., someone is entering or exiting a room



An approach for people counting

- The raw analog output (voltage) patterns is processed by a deep learning model
- A multi-task supervised learning task can simultaneously learn from analog data:
 - the number of people
 - their direction



Yun, J., Kim, D., Kim, D. M., Song, T., & Woo, J. (2023). GAN-based sensor data augmentation: Application for counting moving people and detecting directions using PIR sensors. *Engineering Applications of Artificial Intelligence*, 117, 105508.

References

- [1] Benmansour, A., Bouchachia, A., & Feham, M. (2015). ***Multioccupant activity recognition in pervasive smart home environments***. ACM Computing Surveys (CSUR), 48(3), 1-36.
- [2] Arrotta, L., Bettini, C., Civitaresse, G., & Presotto, R. (2020). ***Context-aware data association for multi-inhabitant sensor-based activity recognition***. In 2020 21st IEEE International Conference on Mobile Data Management (MDM) (pp. 125-130). IEEE.
- [3] Ye, J., Stevenson, G., & Dobson, S. (2015). ***KCAR: A knowledge-driven approach for concurrent activity recognition***. Pervasive and Mobile Computing, 19, 47-70.